

RSMViewer PyMOL Plugin — Manuscript Application Benchmark

Execution times were measured in fresh headless PyMOL sessions, using `time.perf_counter()` around the manuscript workflow commands only. PyMOL startup and ray rendering overhead were excluded so the reported values reflect the actual application steps.

Each workflow was repeated 3 times; the table reports the median runtime and the repeated run values used for the median calculation.

Application	Workflow	Median runtime	Repeated runs	Validated
1	Application 1 — individual motif instance	0.372824 s	0.383155 s, 0.372824 s, 0.369209 s	Yes
2	Application 2 — all motif instances in one family	0.402657 s	0.401385 s, 0.402657 s, 0.408106 s	Yes
3	Application 3 — multi-structure comparison	0.801835 s	0.806097 s, 0.797562 s, 0.801835 s	Yes
4	Application 4 — motif object creation and superposition	0.534849 s	0.521346 s, 0.534849 s, 0.535305 s	Yes
5	Application 5 — TP / FP / FN partitioning	0.406003 s	0.402784 s, 0.407585 s, 0.406003 s	Yes
6	Application 6 — color, combine, and superimpose groups	1.213281 s	1.231832 s, 1.213281 s, 1.144446 s	Yes

Method summary

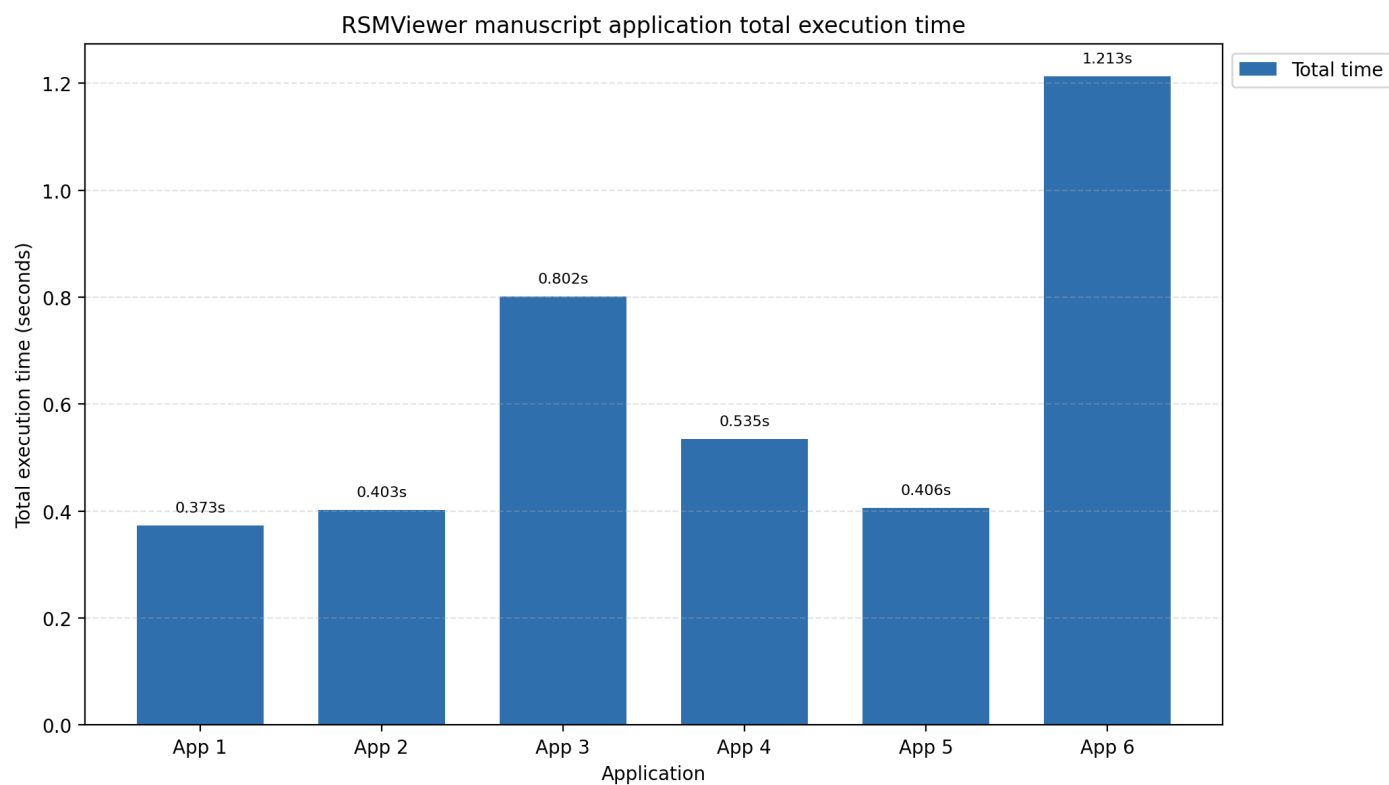
1. Start a fresh PyMOL session in headless mode. 2. Load the plugin and call the same manuscript commands used in the paper. 3. Measure each command with `time.perf_counter()` and accumulate the elapsed time for the full workflow. 4. Repeat the workflow three times and use the median as the reported runtime. 5. Check the resulting motif groups and object counts to ensure the workflow was valid.

Aggregate median across the six applications: 3.731449 s. The fastest routine was Application 2 at 0.372824 s; the slowest was Application 6 at 1.213281 s.

Notes

This benchmark reflects direct command execution in the plugin environment and intentionally excludes PyMOL startup, network fetch overhead, and any ray-traced display generation. The goal is to measure the manuscript workflow itself rather than the full graphical viewer lifecycle.

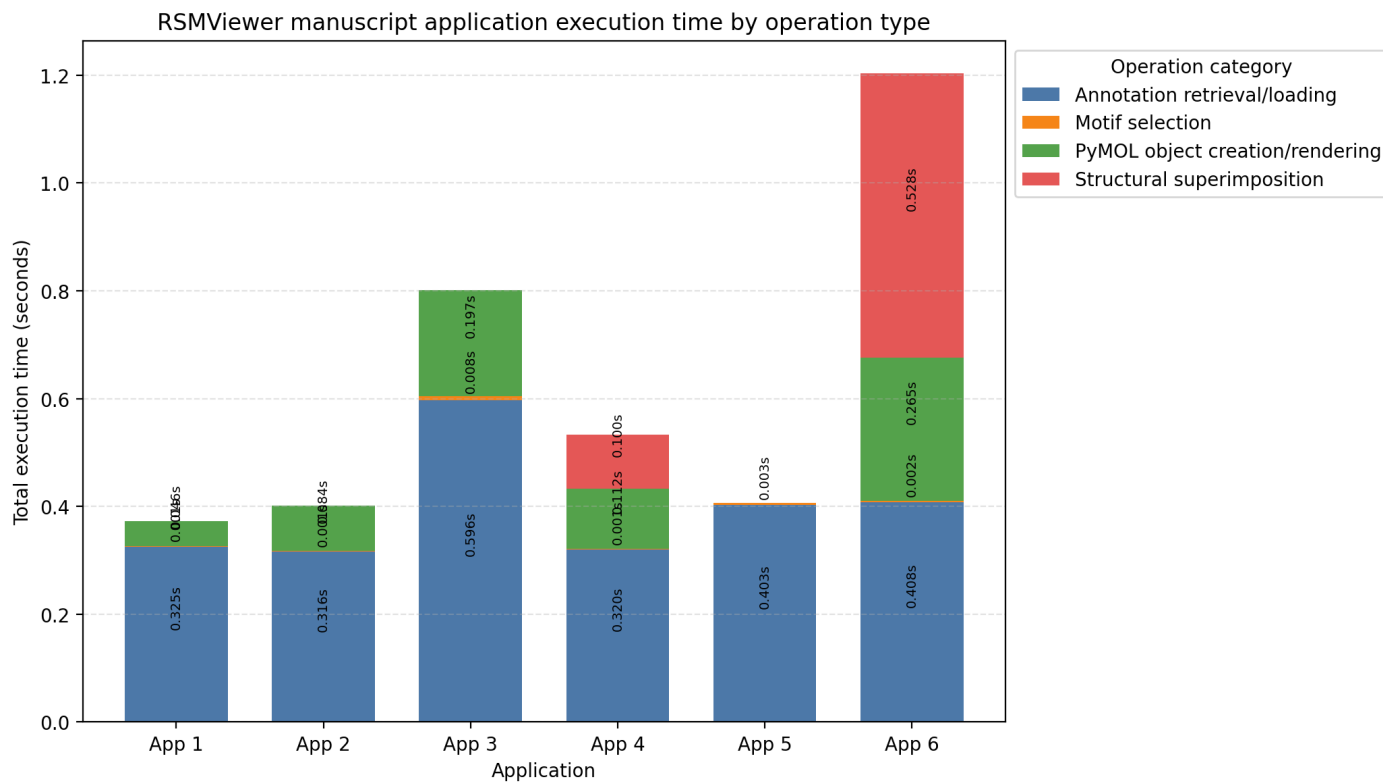
Figure 1 — Total execution time per application



Raw data (median of 3 repeated runs per application):

App	Run 1 (s)	Run 2 (s)	Run 3 (s)	Median (s)	Mean (s)	Min (s)	Max (s)
1	0.383155	0.372824	0.369209	0.372824	0.375063	0.369209	0.383155
2	0.401385	0.402657	0.408106	0.402657	0.404049	0.401385	0.408106
3	0.806097	0.797562	0.801835	0.801835	0.801831	0.797562	0.806097
4	0.521346	0.534849	0.535305	0.534849	0.530500	0.521346	0.535305
5	0.402784	0.407585	0.406003	0.406003	0.405457	0.402784	0.407585
6	1.231832	1.213281	1.144446	1.213281	1.196520	1.144446	1.231832

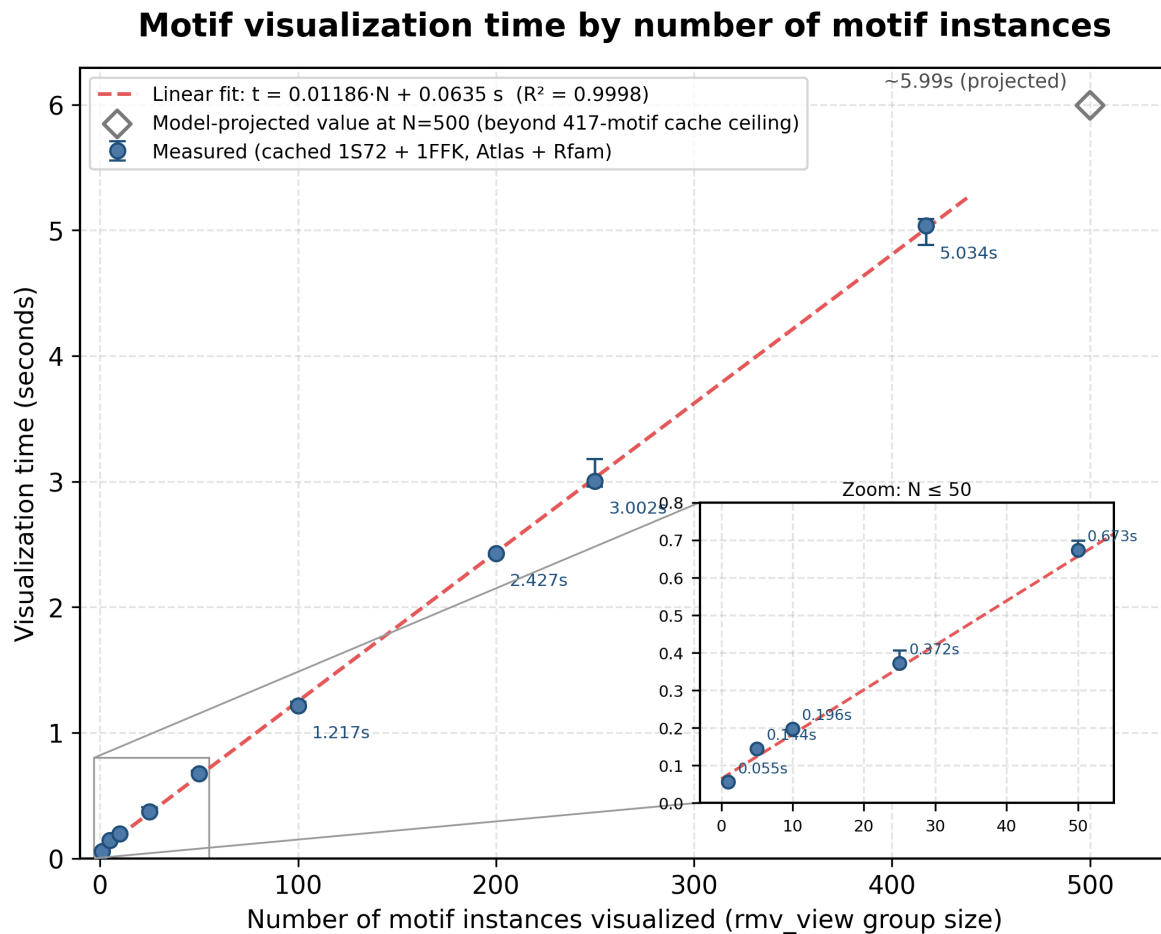
Figure 2 — Execution time by operation category (stacked)



Raw data (median seconds per operation category):

App	Annotation retrieval/loading	Motif selection	PyMOL object creation/rendering	Structural superimposition
1	0.325394	0.001084	0.046342	0.000000
2	0.316327	0.001121	0.084497	0.000000
3	0.596482	0.008060	0.196963	0.000000
4	0.319839	0.001133	0.112430	0.099576
5	0.402890	0.003087	0.000000	0.000000
6	0.408101	0.002162	0.265405	0.528283

Figure 3 — Motif visualization time by number of motif instances



Visualization time = wall-clock duration of the rmv_view command that highlights every residue in the group on the already-loaded structure(s); rmv_fetch and rmv_db are excluded. Motif groups pool 43+ families across 1S72 and 1FFK from cached RNA 3D Motif Atlas + Rfam annotations (no network calls); each point is the median of 3 repeated in-session runs.

Raw data (pool size = 417 cached motif instances across 1S72 + 1FFK, RNA 3D Motif Atlas + Rfam; median of repeated rmv_view runs per group size):

Motif count	Run 1 (s)	Run 2 (s)	Run 3 (s)	Median (s)	Mean (s)	Min (s)	Max (s)	Structures used
1	0.060734	0.053167	0.054981	0.054981	0.056294	0.053167	0.060734	1S72
5	0.145054	0.144409	0.141911	0.144409	0.143791	0.141911	0.145054	1FFK, 1S72
10	0.201826	0.195245	0.196397	0.196397	0.197823	0.195245	0.201826	1FFK, 1S72
25	0.370335	0.371750	0.405999	0.371750	0.382694	0.370335	0.405999	1FFK, 1S72
50	0.697410	0.672662	0.665337	0.672662	0.678470	0.665337	0.697410	1FFK, 1S72
100	1.204635	1.245453	1.216724	1.216724	1.222271	1.204635	1.245453	1FFK, 1S72
200	2.427210	2.405919	2.432204	2.427210	2.421778	2.405919	2.432204	1FFK, 1S72
250	3.179170	3.001825	2.959346	3.001825	3.046780	2.959346	3.179170	1FFK, 1S72
417	4.884354	5.090198	5.034133	5.034133	5.002895	4.884354	5.090198	1FFK, 1S72